Risk Assessment

ACTIVITIY TITLE:																
No.	Hazard Title	Hazard Detail	Causes	Consequences	* Risk Assessment					Existing Controls	Additional Risk Controls Planned/ Required	Residual Risk				
					P	A	E	R	RI			P	A	E	R	RI
Manual Handling																
1	Ergonomic Hazard Physical Hazard	Manually Lifting a heavy load. & Manually lifting a large load lifting a sharp load	• Improper stacking of material • Holding load away from the body • Worke place very narrow	• Musculoskeletal injury • Sprain/strain type of injuries to neck, lower back and joint • Cuts wounds • Crushing injury • Property / equipment damage	3C	2C	-	3C	M	• Training personnel in kinetics of manual lifting. • Ensure load is below the recognized permissible limit. • Use of lifting machines to lift heavy and large loads. • Follow easy lifting method. • Take help if load is large/very heavy. • Use personnel protective equipment as provided. (Safety shoes, coveralls, gloves etc.)	• Ask specific PPE' to be used along with mandatory PPE's	1C	1C	-	1C	LOW
Cutting of Plywood / Timber / gypsum board using circular saw cutting machine																
1	• Physical Hazard • Fire	• Injuries due to moving parts of cutting machine • Electrical shock • High Noise & eye injuries • Dust	• Non competent person • Carelessly handling the machine • Noncomplian ce of proper PPE	• Cuts • Crushing of fingers • Electrocution • Amputations • Burns are usually most severe at the points of contact with the electrical source and the ground	3C	2C		3C	M	• All machines shall be inspected by authorized person prior to use. • RGBY color code system shall be implemented for machines in good and safe condition. • Trained operator shall be allowed to operate the machine. • Operator shall wear double eye protection (safety goggles and	• Safe distance should be maintained at the time of shifting / placing the materials by equipment • The SWL must be indicated on telehandler / JCB never exceeded. • All lifting tools and tackles must be inspected and	1C	1C	1C	1C	LOW

										face shield) during operation. • Operator shall complete daily visual inspection prior to use. • All cable connections shall be inspected and installed by authorized electricians only. Ear Plug shall be use if the noise is more than 80 db. • Electrical connections to the machines must be routed through ELCB. • All leads managed to ensure protection (elevated or covered) • Operator shall be provided to wear ear plug/ ear muff during operation • All machines shall be provided with proper guards to avoid contact with rotating parts. • Operator shall wear double eye protection (safety goggles and face shield) during operation. • Operator shall be provided to wear ear plug/ during operation.	certified including lifting points.					
Painting Works																
1	Physical hazard,	Slip, trip, falls, dropping paint container, inhalation of fumes may lead to breathing	Not using the proper PPE, Usage of damaged work platform, usage of damaged	Fall from height, skin irritation, damage to lungs, falling of container from height leading to injury of people working	3C	2C		3C	M	• PTW system properly implemented, MSDS properly explained to workers, TBT done before start of work, all	Strict adherence of PPE will be ensured	1C	1C	-	1C	LOW

		problems, skin allergy and spillage	containers, poor house keeping							workers provided with complete PPE.						
Use of Power Hand Tools																
1	Electrical Hazard Fire Hazard Physical Hazard	• Short Circuit • Use of portable power tools • Use of Hand tools	• Non competent untrained Workes • Usage of damaged or improper Hand and power tools • Non isolation of utilities	• Injury to personnel • Electrocution • Cuts • Burns • Amputation • Fire	3C	2C	-	3C	M	• All portable power tools shall be fitted with adequate fuse protection either in the tool body or in its electrical plug. • Tools shall be maintained in good working condition and tested at regular intervals.	• Circuits of all Electrical Equipment will be protected by 30mA ELCB • Disc/Wheel/Tool Kit shall be changed only after tool has been disconnected from power source	1C	1C	-	1C	LOW
LV Cable Laying, Dressing and Termination Work																
1	Physical Hazard Heavy Load Ergonomic Hazard Physical Hazard	• Collapse/ rollover of cable drum • Failure of Cable Drum Stand • Use of Hand tools • Equipment failure	• Untrained non competent workers • Working without isolating power	• Electrocution • Property / equipment damage • Cuts, wounds • Burns	3C	2C	-	3C	M	• Training personnel in kinetics of manual lifting. • Ensure load is below the recognized permissible limit. • Use of lifting machines to lift heavy and large loads. • Follow easy lifting method. • Take help if load is large/very heavy. • Use personnel protective equipment as provided. (Safety shoes, coveralls, gloves etc.) • All portable power tools shall be fitted with adequate fuse protection either in the tool body or in its electrical plug.	• Circuits of all Electrical Equipment will be protected by 30mA ELCB • No work on live circuit • Implement LOTO (lock out tag out system) • Use insulated gloves • Keep away unauthorized workers during cable testing and Megara work • Keep away all sharp tools such as knife and sharp blades from the electrical cables	1C	1C	-	1C	LOW

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Work at Heights on Platforms																
1	Physical Hazard	Work at Hight Improper Access / limited access in and out Incomplete scaffold platform / Incomplete edge Wrong modifications to scaffold • platform	Lack of training Overloading of platform Not following site rules & Regulation Invisibility Adverse weather conditions Adjacent under work activities Poor maintenance	• Falling of personnel from height • Injury to personnel • Falling of hand tools objects from height	3C	2C	-	3C	M	1- PTW available and strictly followed. 2- Tool box talk and JSA carried out before commencing the job for each shift. 3- Supervisor / Foreman shall be made available at all times during work. 4- Scaffolding platform inspected / approved by competent person and tagged (Green) safe for use. 5- Work At Height training must be completed and procedures shall be complied with. 6- Scaffolding must not be over loaded at any time. Display load bearing capacity on scaffold. 7- Wrist strap must be issued and used by personnel working at heights. 8- Use tool box or collection baskets for materials handling on top of the scaffold to avoid rolling or falling	Use competent and trained Workers. □ Internal training shall be provided to the workers.	1C	1C	1C	1C	LOW

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											9. All hand tools must be secured properly to prevent from falling at height. 10- Exclusion zone must be maintained underneath working at height location. No one is allowed to enter or walk inside this zone. • 11- Sufficient warning signs						

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Risk Assessment Steps:
Step 1: Identify Hazards Title (e.g. Physical Hazard, Fire Hazard, and Electrical Hazards).
Step 2: List Hazards Details identified (e.g. Tripping Hazard, Flammable Materials stored, Working in Electrical Cabinets).
Step 3: List Causes of Hazard (e.g. Untidy workplace, Hot work, un-isolated electrical supply).
Step 4: List Consequences of Hazard (e.g. Personnel Injury, Fire, electric shock).
Step 5: Determine Likelihood and Consequences to People, Assets, Environment, and Reputation from attached Risk Assessment Matrix (Page 2) (this should be the most probable outcome without controls and should be listed as 2A, 3D etc.).
Step 6. Calculate the Risk Index using the Qualitative or Semi-Quantitative Risk Assessment Matrix as deemed appropriate. This is a product of Likelihood and Consequence and should be stated Low, Medium, High or Very High).
Step 7. List Existing Controls. (e.g. Tidy Site Policy, Hot Work Permit, Isolation of Electrical Supply).
Step 8. List Safety Critical Activities and Additional Controls that may be deemed necessary to reduce risk level to ALARP.
Step 9: Recalculate Likelihood and Consequences to People, Assets, Environment, and Reputation from attached Risk Assessment Matrix.
Step 10. Recalculate the Risk Index.
Step 11. If residual risk deemed to be High or Very High then risk must be reduced further before proceeding, if risk is medium then note that risk reduction must be planned, monitored and documented, if risk is low then risk is acceptable without required further action.

Likelihood	Descriptor to assist in Qualitative Assessment of Likelihood/ Frequency of Occurrence.
A	Has occurred within the industry worldwide.
B	Has occurred regionally within the industry or judged likely to have occurred.
C	Has occurred in the industry within UAE or judged likely to have occurred.
D	Has occurred in the industry within UAE on an annual basis or judged likely to have occurred.
E	May occur in the industry multiple times annually.

* Risk Assessment Matrix – Qualitative

					LIKELIHOOD				
Consequences	P= People	A= Assets	E= Environment	R= Reputation	A	B	C	D	E
5- Catastrophic	Multiple fatalities	Extensive damage	Massive effect	International impact					VERY HIGH
4 - Severe	Single fatality or permanent disability	Major damage	Major effect	National impact			HIGH		
3 – Critical	Major injury or health effects	Localized effect	Localized effect	Considerable impact		MEDIUM			
2 – Marginal	Minor injury or health effects	Minor effect	Minor effect	Minor impact					
1 - Negligible	Slight injury or health effects	Slight effect	Slight effect	Slight impact	LOW				

ALARP